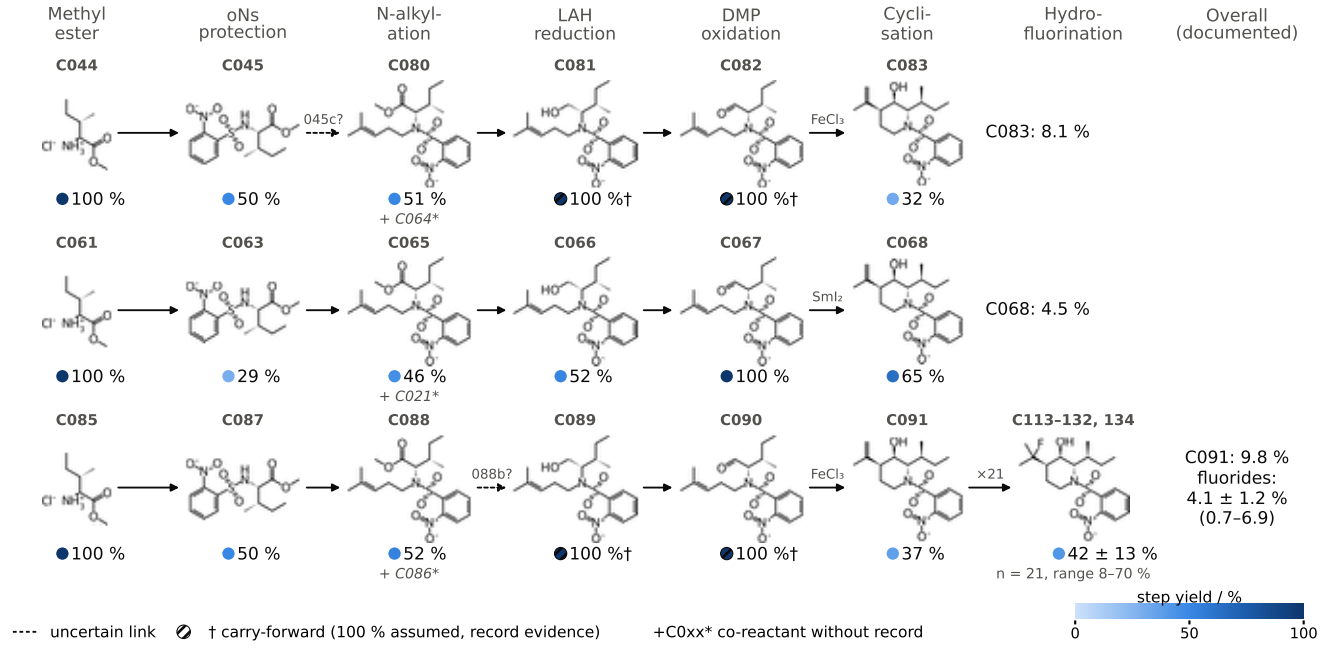

Methyl ester
oNs protection
N-alkyl-ation
LAH reduction
DMP oxidation
Cycli-sation
Hydro-fluorination
Overall (documented)
C044
C045
C080
C081
C082
C083
045c?
FeCl₃
C083: 8.1 %
100 %
50 %
51 %
+ C064*
100 %†
100 %†
32 %
C061
C063
C065
C066
C067
C068
SmI₂
C068: 4.5 %
100 %
29 %
46 %
+ C021*
52 %
100 %
65 %
C085
C087
C088
C089
C090
C091
C113–132, 134
088b?
FeCl₃
×21
C091: 9.8 %
fluorides:
4.1 ± 1.2 %
(0.7–6.9)
100 %
50 %
52 %
+ C086*
100 %†
100 %†
37 %
42 ± 13 %
n = 21, range 8–70 %
step yield / %
0
50
100
uncertain link
† carry-forward (100 % assumed, record evidence)
+C0xx* co-reactant without record